

Alternative: plain switches + Tailscale on generic Linux, no GL-iNet routers

Same theatre topology (isolated /24 per theatre, subnet routing, Tailscale ACLs doing cross-theatre isolation) — different hardware. Instead of a GL-iNet Slate AX doing router+NAT+DHCP+Tailscale in one appliance, use a plain unmanaged switch for LAN fanout plus a small generic Linux box (mini-PC/NUC, or the theatre's existing control laptop) running Tailscale as a subnet router with NAT/DHCP built from standard Linux tools (nftables + dnsmasq). Config in config/alt-tailscale-switch/.

Why a router-equivalent device is still required, not eliminated: the ATEM and BirdDog Play are closed appliances — they can't run the Tailscale client themselves. Some device must still advertise the theatre's subnet on their behalf. "Pure Tailscale, no router" only works if literally every device can run Tailscale; since two can't, this alternative just moves the subnet-router role onto generic Linux instead of GL-iNet's firmware. The tailscale up invocations themselves are identical either way — see config/tailscale-up-all-devices.sh — Tailscale doesn't care what hardware runs it.

Comparison

	GL-iNet Slate AX (chosen)	Generic Linux box + switch (alternative)
Setup effort	Config file, done (see config/gl-inet/)	Build + harden NAT/DHCP/Tailscale from scratch per box
Throughput confidence	Vendor datasheet, ~550 Mbps — but that is kernel WireGuard; Tailscale's userspace wireguard-go is estimated at ~150-250 Mbps combined on the Slate AX, unbenchmarked (see docs/bandwidth-analysis.md). The previous choice, the A-1300, was dropped on 2026-10-06 for exactly this gap (170 Mbps kernel vs ~30-70 Mbps Tailscale est.)	Unverified until benchmarked on whatever's chosen (an x86 mini-PC will very likely beat any travel router under Tailscale)
Physical footprint	One compact unit per theatre	Separate PC + separate switch + 2 power supplies
Dedicated ATEM port	Built-in (2 LAN ports)	Needs a 2nd NIC/dongle to replicate
Wi-Fi	Built-in	Extra hardware if needed at all
Spares/swap-in	Re-flash identical UCI config	Re-provision a whole OS image
Maintenance	Vendor firmware updates	Ongoing OS patching, our responsibility
Flexibility/CPU headroom	Fixed, embedded-class	Higher, if it matters later
Unit cost at required throughput	~\$120 travel router (Slate AX list)	Often more, once a real NIC + case + PSU are added

Recommendation: GL-iNet

This is 12 identical small router/switch appliances for a touring event that need to just work — exactly the product category a travel router is built for. A generic Linux box gives more flexibility and CPU headroom we don't need here, at the cost of more setup work, more failure modes, and a bulkier kit per theatre. (A travel router's throughput "guarantee" is weaker than it looks under Tailscale — that is why the A-1300 was replaced by the Slate AX — and if the Slate AX benchmark in `docs/bandwidth-analysis.md` comes back low too, CPU headroom stops being something "we don't need" and this comparison should be revisited.) The one real advantage of the alternative — no vendor lock-in — doesn't outweigh those costs for this deployment. Keep GL-iNet as the primary; this alternative is documented for completeness, not as an equally-weighted option.